

Memory Bytes, Words, and Addresses

Basic Idea

- A **byte** = 8 bits (smallest unit of data stored in memory).
- A **word** = 2 bytes = 16 bits (used by 16-bit microprocessors).
- Every **byte** has its own **unique address**.
- When two bytes form one word, the **lower address** of those two bytes is considered the **word address**.

Example: Two Words in Memory

Memory Address	Data (in bits)
0	01000001
1	01000010
2	01000011
3	01000100
4	01000101
5	01000110
6	01000111
7	01001000

Word Formation

Word	Bytes Used	Combined (16-bit)	Word Address
Word 1	Byte 0 + Byte 1	0100000101000010	0
Word 2	Byte 2 + Byte 3	0100001101000100	2
Word 3	Byte 4 + Byte 5	0100010101000110	4
Word 4	Byte 6 + Byte 7	0100011101001000	6

Explanation

Each word is made up of two consecutive bytes.

For example:

- **Word 1** → formed by bytes at address **0 and 1**. Word address = **0**.
- **Word 2** → formed by bytes at address **2 and 3**. Word address = **2**.

This means if a program says, "Fetch the word stored at memory address 2", the microprocessor will read both bytes 2 and 3 together.

In short

Term	Meaning
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Byte	8 bits, smallest memory unit
Word	16 bits (2 bytes)
Word Address	The address of the first byte (lower address)
Example	Word at address 2 = bytes at 2 and 3